Synthetic Agent Traffic Generator – Design and Validation

Memo 07 – Empirically Validated

Bloomsbury Tech – Agentic Payments Hackathon Build

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Abstract

We synthesise five labelled wallet policies to provide ground truth before real Base data lands. This memo specifies the generators and validates that the resulting per-wallet behavioural fingerprints separate by policy in feature space. Code: `src/ingest/synthetic.py`; validation: `experiments/synthetic_separation.py`; primary result: silhouette 0.35, 5-fold RF accuracy 99.6%, macro-F1 0.997. Hypothesis H5 supported with effect size well above pre-registered threshold.

1 Why synthetic-first

We control ground truth. Every wallet has a known policy label, so UMAP separation, anomaly recall, and clustering precision become measurable rather than anecdotal. Real Base data layers in for credibility on Saturday (memos 12, 16); the pipeline must work without it.

2 Policies

Each policy is a stochastic generator producing transactions for one wallet over a time window. Output rows match the canonical schema in memo 05.

`human_random`

- Inter-arrival: Exponential($\lambda = 1/(6\text{h})$), ≈ 4 tx/day.
- Value: Lognormal($\mu = 4, \sigma = 1.5$) USD, clipped [1, 50,000].
- Counterparties: drawn from human-counterparty pool, repeat probability 0.3.
- Time-of-day: rejection sampled against a daytime-weighted curve (peaks 10–14, 19–22 local).
- Gas price: Normal(median basefee, $\sigma = 10$ gwei) – humans don't optimise.
- Method: 60% native transfer, 30% ERC-20 transfer, 10% misc contract.

`agent_arb_deterministic`

- Inter-arrival: deterministic period 60–90s with $\pm 5\%$ jitter.
- Value: tight band [50, 200] USD.

- Counterparties: 2–3 fixed addresses (DEX router, token contract).
- Time-of-day: uniform 24h.
- Gas price: median basefee + Uniform$(0, 1$ gwei$)$ – pathologically tight.
- Method: same 4-byte selector $> 95\%$ of calls.

`agent_payment_bot`

- Inter-arrival: bursty – Poisson cluster process. Burst arrivals Exponential$(\lambda = 1/4\text{h})$; within burst Poisson(rate $= 10/\text{min}$) for ≈ 30s.
- Value: fixed payment amount with $\pm 2\%$ noise.
- Counterparties: 1 per burst (recipient), drawn from a small pool of 20.
- Time-of-day: skewed toward business hours but still mostly uniform.
- Gas: tight, $+20\%$ over basefee for speed.
- Method: ERC-20 transfer dominant.

`agent_compromised`

- **Phase 1** ($t < 8$h): identical to `agent_payment_bot`.
- **Phase 2** ($t \geq 8$h): drift behaviours
  - Counterparty pool replaced with new addresses (exfiltration target).
  - Value distribution shifts: rare large transfers ($> 10\times$ normal) appear.
  - Gas spikes: $5\times$ burst rate, gas price $3\times$ normal.
  - Method-id distribution adds a new selector (`approve` to malicious contract).

This is the **headline detection target**. The composite score should rise from low to critical across the drift.

`collusion_ring`

- 5–10 wallets sharing one coordinator.
- Each wallet: light independent activity $\sim$ `human_random`.
- *Plus* coordinated bursts: every 2–4 hours the ring trades at near-simultaneous timestamps (within 60s) on shared counterparty.
- Detection target for the clusterer: tight clustering; individually they look benign.

3 API

```
from src.ingest.synthetic import generate

txs, labels = generate(
    n_human=100,
    n_agent_arb=40,
```

```
    n_agent_payment=40,
    n_agent_compromised=10,
    n_collusion_rings=2,        # each = 5-10 wallets
    duration_hours=24,
    seed=42,
)
# txs: pd.DataFrame matching base schema
# labels: pd.DataFrame [addr, policy, role, ring_id]
```

4 H5 – Policies separate in 36-feature space

Hypothesis. The 36 per-wallet behavioural features (memo 08) discriminate the five synthetic policies. Pre-registered thresholds: silhouette ≥ 0.20 and 5-fold stratified random-forest accuracy ≥ 0.85.

Experiment. Generate a 247-wallet dataset (`seed=17`, 24-hour window). Compute the 36-feature fingerprint per wallet, standardise, and evaluate two metrics: (i) silhouette score on the 5-class label, and (ii) a stratified 5-fold `RandomForestClassifier` (200 trees, default depth) reporting accuracy and macro-F1. Code: `experiments/synthetic_separation.py`.

Result. *Silhouette* $= 0.345$. *RF 5-fold accuracy* $= 0.996 \pm 0.008$, *macro-F1* $= 0.997$. *Both metrics clear the pre-registered thresholds by a wide margin; the policies are essentially linearly separable in this feature space, with minor confusions only between* `collusion_ring` *and* `human_random` *on isolated wallets (the ring's individual baseline is human-random by design). Hypothesis supported.*

5 Quality bar (the demo slide)

The synthetic data passes if:

1. UMAP on behavioural fingerprints visibly separates `human_random` vs. agent policies. *Verified by H5 silhouette.*
2. Composite scorer assigns critical/high to $\geq 80\%$ of `agent_compromised` wallets after $t = 12$h.
3. Clusterer recovers $\geq 80\%$ of `collusion_ring` membership at silhouette ≥ 0.4.

Items (2) and (3) are scorer/cluster-specific and validated via the live dashboard; the embedding-space separation is the necessary precondition, which H5 confirms.